

$$\frac{dv}{dt} = g - \frac{c}{m} v$$

$$\frac{dv}{dt} + \frac{c}{m} v = g$$

$$u(t) = e^{\int \frac{c}{m} dt} = e^{\frac{c}{m} t}$$

$$e^{\frac{c}{m} t} \left(\frac{dv}{dt} + \frac{c}{m} v \right) = e^{\frac{c}{m} t} g$$

$$\frac{d}{dt} (v e^{\frac{c}{m} t}) = e^{\frac{c}{m} t} g$$

$$v e^{\frac{c}{m} t} = \int e^{\frac{c}{m} t} g dt$$

$$v e^{\frac{c}{m} t} = g \frac{m}{c} e^{\frac{c}{m} t} + k$$

$$v(t) = g \frac{m}{c} + e^{-\frac{c}{m} t} k$$

$$v(t) = 0$$

$$v(0) = g \frac{m}{c} + k = 0 \Rightarrow k = -g \frac{m}{c}$$

$$v(t) = \frac{g m}{c} \left(1 - e^{-\frac{c}{m} t} \right)$$